EXAMPLE 4.6 Obtain the direct form-I, direct form-II, cascade and parallel form realizations of the LTI system governed by the equation

$$y(n) = -\frac{13}{12}y(n-1) - \frac{9}{24}y(n-2) - \frac{1}{24}y(n-3) + x(n) + 4x(n-1) + 3x(n-2)$$

Solution:

Direct form-I

$$\text{Given } y(n) = -\frac{13}{12}y(n-1) - \frac{9}{24}y(n-2) - \frac{1}{24}y(n-3) + x(n) + 4x(n-1) + 3x(n-2)$$

Taking Z-transform on both sides, we get

$$Y(z) = -\frac{13}{12}z^{-1}Y(z) - \frac{9}{24}z^{-2}Y(z) - \frac{1}{24}z^{-3}Y(z) + X(z) + 4z^{-1}X(z) + 3z^{-2}X(z)$$

The direct form-I structure can be obtained from the above equation as shown in Figure 4.22.

Direct form-II

Taking Z-transform of the given difference equation, we have

$$Y(z) = -\frac{13}{12}z^{-1}Y(z) - \frac{9}{24}z^{-2}Y(z) - \frac{1}{24}z^{-3}Y(z) + X(z) + 4z^{-1}X(z) + 3z^{-2}X(z)$$

i.e. $$Y(z)+\frac{13}{12}z^{-1}Y(z)+\frac{9}{24}z^{-2}Y(z)+\frac{1}{24}z^{-3}Y(z)=X(z)+4z^{-1}X(z)+3z^{-2}X(z)$$

i.e. $$Y(z)\left[1+\frac{13}{12}z^{-1}+\frac{9}{24}z^{-2}+\frac{1}{24}z^{-3}\right]=X(z)\,[1+4z^{-1}+3z^{-2}]$$

Therefore, the transfer function of the system is

$$H(z)=\frac{Y(z)}{X(z)}=\frac{1+4z^{-1}+3z^{-2}}{1+(13/12)\,z^{-1}+(9/24)\,z^{-2}+(1/24)\,z^{-3}}$$

Let $$\frac{Y(z)}{X(z)}=\frac{Y(z)}{W(z)}\,\frac{W(z)}{X(z)}$$

where $$\frac{W(z)}{X(z)}=\frac{1}{1+(13/12)\,z^{-1}+(9/24)\,z^{-2}+(1/24)\,z^{-3}}$$

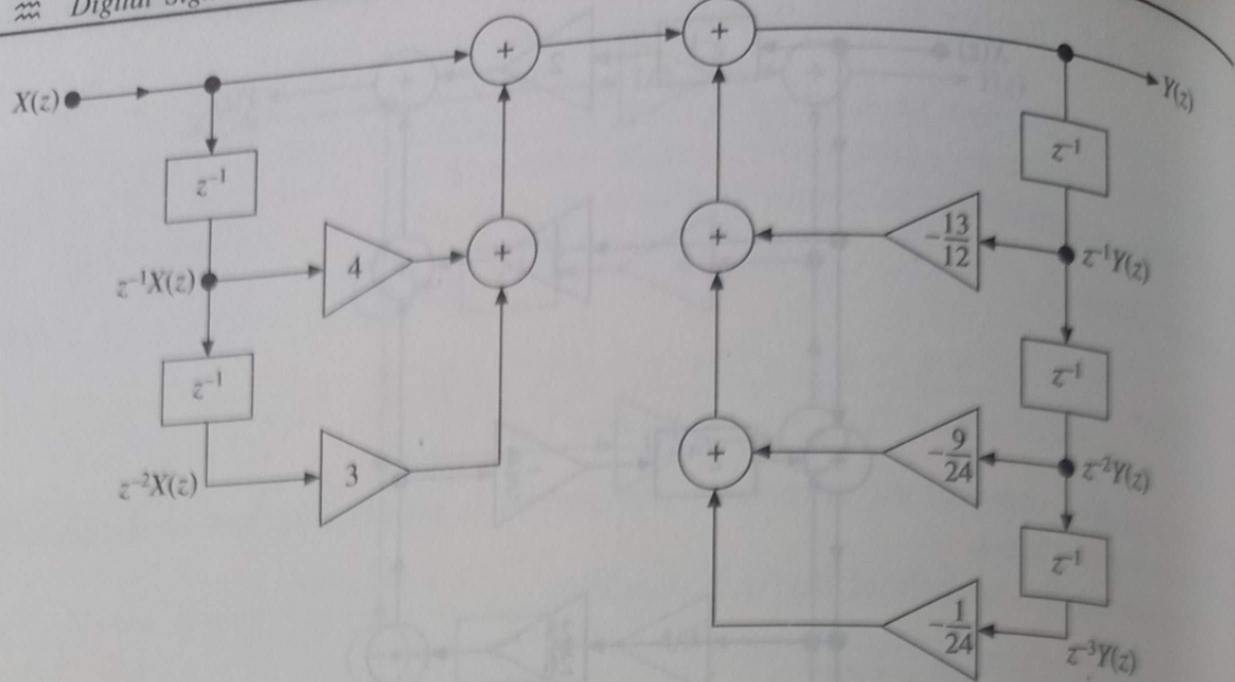

Figure 4.22 Direct form-I realization structure (Example 4.6).

and $$\frac{Y(z)}{W(z)} = 1 + 4z^{-1} + 3z^{-2}$$

On cross multiplying the above equations, we get

$$W(z)\left[1 + \frac{13}{12}z^{-1} + \frac{9}{24}z^{-2} + \frac{1}{24}z^{-3}\right] = X(z)$$

i.e. $$W(z) = X(z) - \frac{13}{12}z^{-1}W(z) - \frac{9}{24}z^{-2}W(z) - \frac{1}{24}z^{-3}W(z)$$

and $$Y(z) = W(z) + 4z^{-1}W(z) + 3z^{-2}W(z)$$

The above equations for $W(z)$ and $Y(z)$ can be realized by a direct form-II structure as shown in Figure 4.23.

Cascade form

The transfer function is $H(z) = \dfrac{Y(z)}{X(z)} = \dfrac{1 + 4z^{-1} + 3z^{-2}}{1 + (13/12)z^{-1} + (9/24)z^{-2} + (1/24)z^{-3}}$

Factorizing the numerator and denominator, we have

$$H(z) = \frac{(1 + z^{-1})(1 + 3z^{-1})}{[1 + (1/2)z^{-1}][1 + (1/3)z^{-1}][1 + (1/4)z^{-1}]}$$

Since there are three first order factors in the denominator of $H(z)$, $H(z)$ can be expressed as product of three sections.

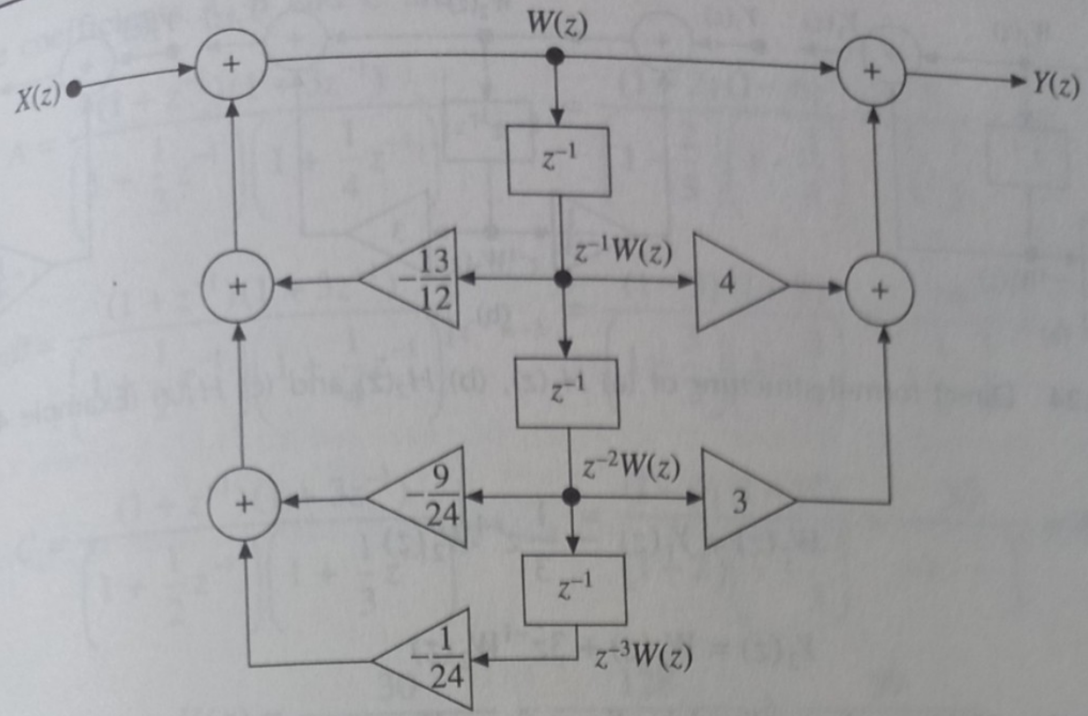

Figure 4.23 Direct form-II realization structure (Example 4.6).

Let

$$H(z) = H_1(z) \;\; H_2(z) \;\; H_3(z)$$

where $$H_1(z) = \frac{1+z^{-1}}{1+(1/2)\,z^{-1}},\; H_2(z) = \frac{1+3z^{-1}}{1+(1/3)z^{-1}} \text{ and } H_3(z) = \frac{1}{1+(1/4)z^{-1}}$$

The transfer function $H_1(z)$ can be realized in direct form-II structure as shown in Figure 4.24(a).

Let $$H_1(z) = \frac{Y_1(z)}{X(z)} = \frac{Y_1(z)}{W_1(z)} \cdot \frac{W_1(z)}{X(z)} = \frac{1+z^{-1}}{1+(1/2)z^{-1}}$$

where $$\frac{W_1(z)}{X(z)} = \frac{1}{1+(1/2)\,z^{-1}} \text{ and } \frac{Y_1(z)}{W_1(z)} = 1+z^{-1}$$

∴ $$W_1(z) = X(z) - \frac{1}{2}z^{-1}W_1(z)$$

and $$Y_1(z) = W_1(z) + z^{-1}W_1(z)$$

The transfer function $H_2(z)$ can be realized in direct form-II structure as shown in Figure 4.24(b).

Let $$H_2(z) = \frac{Y_2(z)}{Y_1(z)} = \frac{Y_2(z)}{W_2(z)} \cdot \frac{W_2(z)}{Y_1(z)} = \frac{1+3z^{-1}}{1+(1/3)z^{-1}}$$

where $$\frac{W_2(z)}{Y_1(z)} = \frac{1}{1+(1/3)\,z^{-1}} \text{ and } \frac{Y_2(z)}{W_2(z)} = 1+3z^{-1}$$

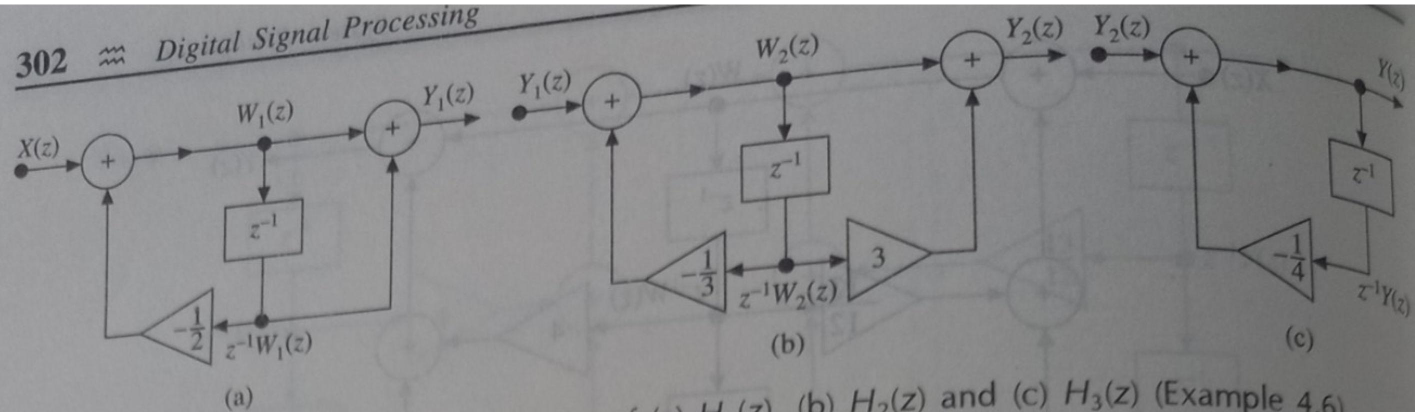

Figure 4.24 Direct form-II structure of (a) $H_1(z)$, (b) $H_2(z)$ and (c) $H_3(z)$ (Example 4.6).

$$\therefore \qquad W_2(z) = Y_1(z) - \frac{1}{3}z^{-1}W_2(z)$$

and

$$Y_2(z) = W_2(z) + 3z^{-1}W_2(z)$$

The transfer function $H_3(z)$ can be realized in direct form-II structure as shown in Figure 4.24(c).

Let

$$H_3(z) = \frac{Y(z)}{Y_2(z)} = \frac{1}{1+(1/4)z^{-1}}$$

$$\therefore \qquad Y(z) = Y_2(z) - \frac{1}{4}z^{-1}Y(z)$$

The cascade structure of the given system is obtained by connecting the individual sections shown in Figures 4.24(a), (b) and (c) in cascade as shown in Figure 4.25.

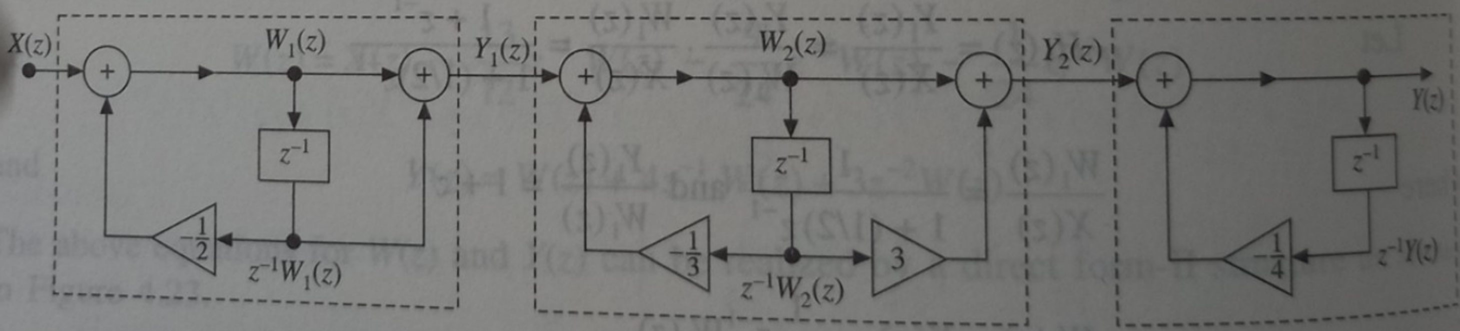

Figure 4.25 Cascade realization of the system (Example 4.6).

Parallel form

Consider the equation $H(z) = \dfrac{(1+z^{-1})(1+3z^{-1})}{[1+(1/2)z^{-1}][1+(1/3)z^{-1}][1+(1/4)z^{-1}]}$

By partial fraction expansion, we have

$$H(z) = \frac{A}{1+(1/2)z^{-1}} + \frac{B}{1+(1/3)z^{-1}} + \frac{C}{1+(1/4)z^{-1}}$$

where, the coefficients A, B and C are

$$A = \frac{(1+z^{-1})(1+3z^{-1})}{\left(1+\frac{1}{3}z^{-1}\right)\left(1+\frac{1}{4}z^{-1}\right)}\Bigg|_{z^{-1}=-2} = \frac{(1-2)(1-6)}{\left(1-\frac{2}{3}\right)\left(1-\frac{2}{4}\right)} = \frac{5}{\frac{1}{3}\cdot\frac{1}{2}} = 30$$

$$B = \frac{(1+z^{-1})(1+3z^{-1})}{\left(1+\frac{1}{2}z^{-1}\right)\left(1+\frac{1}{4}z^{-1}\right)}\Bigg|_{z^{-1}=-3} = \frac{(1-3)(1-9)}{\left(1-\frac{3}{2}\right)\left(1-\frac{3}{4}\right)} = \frac{16}{-\frac{1}{2}\cdot\frac{1}{4}} = -128$$

$$C = \frac{(1+z^{-1})(1+3z^{-1})}{\left(1+\frac{1}{2}z^{-1}\right)\left(1+\frac{1}{3}z^{-1}\right)}\Bigg|_{z^{-1}=-4} = \frac{(1-4)(1-12)}{(1-2)\left(1-\frac{4}{3}\right)} = \frac{33}{-1\cdot-\frac{1}{3}} = 99$$

$\therefore$
$$H(z) = \frac{30}{1+(1/2)z^{-1}} - \frac{128}{1+(1/3)z^{-1}} + \frac{99}{1+(1/4)z^{-1}}$$

Let
$$H(z) = \frac{Y(z)}{X(z)}$$

$\therefore$
$$\frac{Y(z)}{X(z)} = \frac{30}{1+(1/2)z^{-1}} - \frac{128}{1+(1/3)z^{-1}} + \frac{99}{1+(1/3)z^{-1}}$$

$\therefore$
$$Y(z) = \frac{30}{1+(1/2)z^{-1}}X(z) - \frac{128}{1+(1/3)z^{-1}}X(z) + \frac{99}{1+(1/4)z^{-1}}X(z)$$

Let $\quad Y(z) = Y_1(z) + Y_2(z) + Y_3(z)$

where $Y_1(z) = \frac{30}{1+(1/2)z^{-1}}X(z)$; $Y_2(z) = -\frac{128}{1+(1/3)z^{-1}}X(z)$ and $Y_3(z) = \frac{99}{1+(1/4)z^{-1}}X(z)$

Let $\quad H_1(z) = \frac{Y_1(z)}{X(z)} = \frac{30}{1+(1/2)z^{-1}}$; $H_2(z) = \frac{Y_2(z)}{X(z)} = -\frac{128}{1+(1/3)z^{-1}}$

and
$$H_3(z) = \frac{Y_3(z)}{X(z)} = \frac{99}{1+(1/4)z^{-1}}$$

The transfer function $H_1(z)$ can be realized in direct form-I structure as shown in Figure 4.26(a).

Let
$$H_1(z) = \frac{Y_1(z)}{X(z)} = \frac{30}{1+(1/2)z^{-1}}$$

On cross multiplying and rearranging, we get

$$Y_1(z) = 30X(z) - \frac{1}{2}z^{-1}Y(z)$$

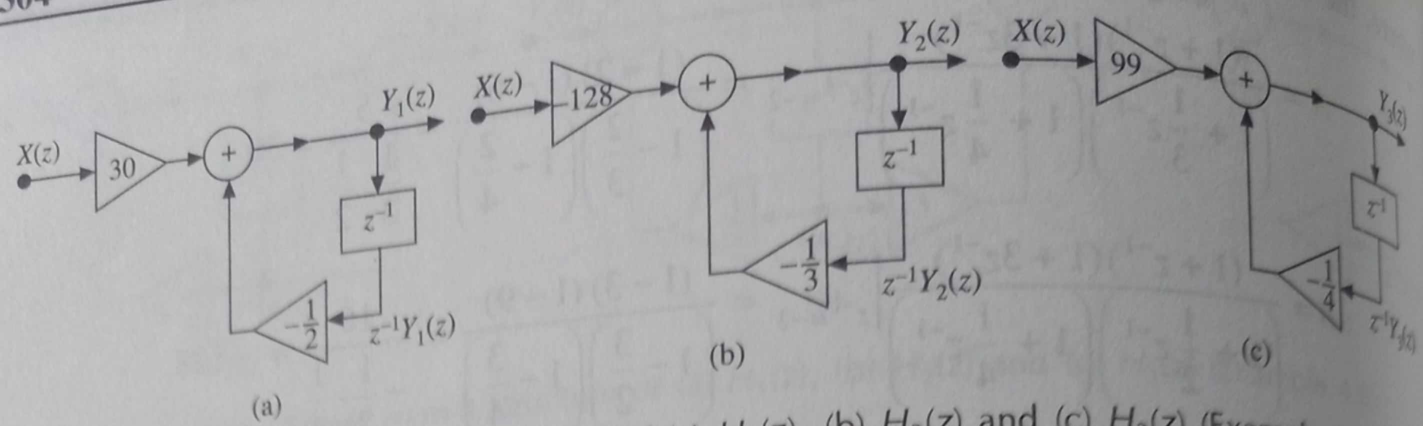

Figure 4.26 Direct form-I structure of (a) $H_1(z)$, (b) $H_2(z)$ and (c) $H_3(z)$ (Example 4.6).

The transfer function $H_2(z)$ can be realized in direct form-I structure as shown in Figure 4.26(b).

Let
$$H_2(z) = \frac{Y_2(z)}{X(z)} = -\frac{128}{1 + (1/3)\,z^{-1}}$$

On cross multiplying and rearranging, we get

$$Y_2(z) = -128\,X(z) - \frac{1}{3}z^{-1}Y_2(z)$$

The transfer function $H_3(z)$ can be realized in direct form-I structure as shown in Figure 4.26(c).

Let
$$H_3(z) = \frac{Y_3(z)}{X(z)} = \frac{99}{1 + (1/4)\,z^{-1}}$$

On cross multiplying and rearranging, we get

$$Y_3(z) = 99\,X(z) - \frac{1}{4}z^{-1}Y_3(z)$$

The overall structure is obtained by connecting the individual sections in Figures 4.26(a), (b) and (c) in parallel as shown in Figure 4.27.

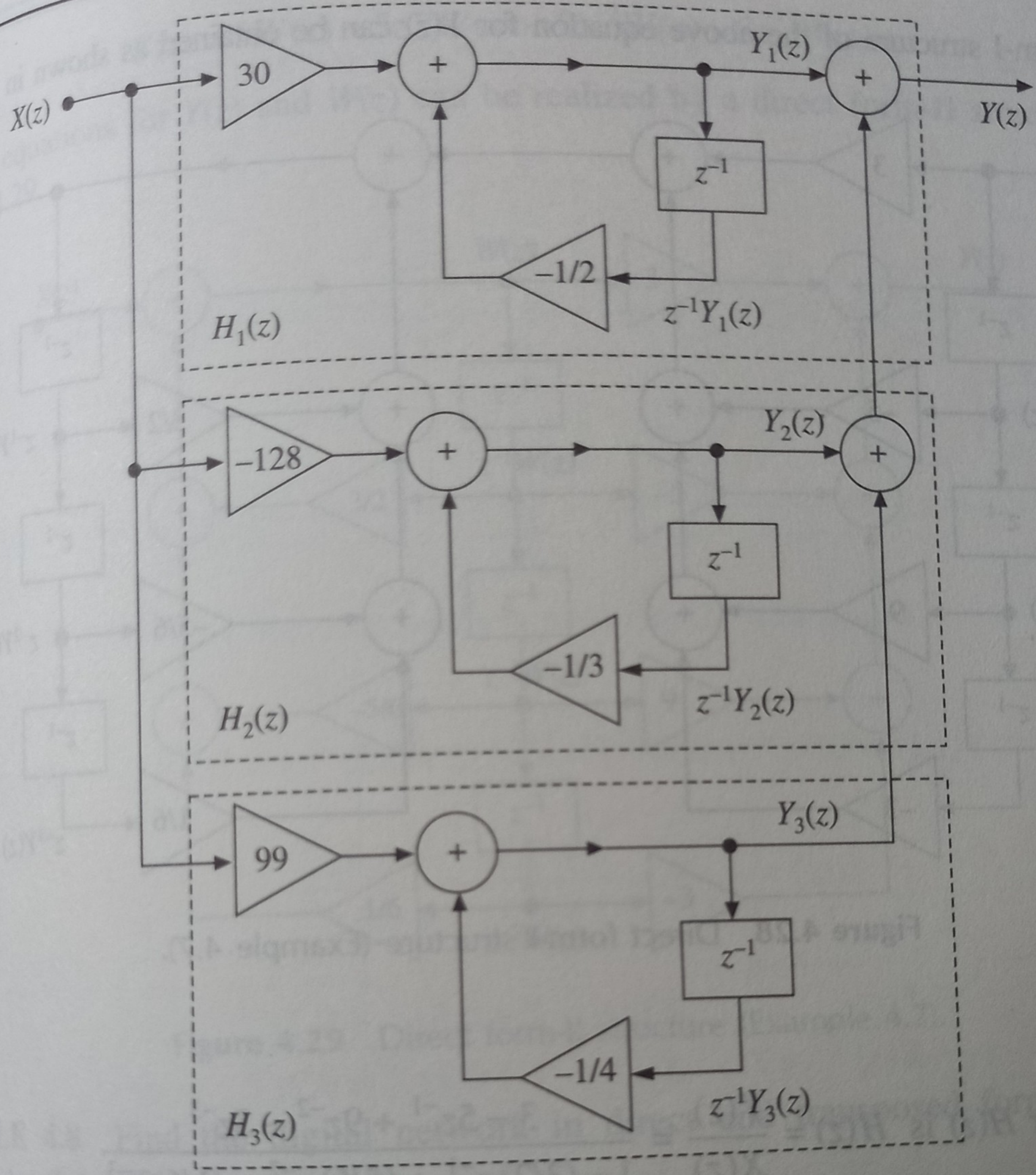

Figure 4.27 Parallel form realization (Example 4.6).